

DRAFT

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Question #: 1

A researcher is studying genetic disorders caused by connective tissue defects and is focusing on the covalent cross-links that are found in collagen and elastin. These cross-links increase the stability and strength of connective tissue. The cross-linking reaction occurs when specific amino acid residues are post-translationally modified in these proteins. Which of the following amino acids most likely undergoes this process?

- A. Glycine
- B. Lysine
- C. Proline
- D. Alanine
- E. Cysteine

Question #: 2

An investigator is studying components of connective tissue. A biopsy from a healthy volunteer shows the proteoglycan aggregates isolated from the extracellular matrix. Which of the following best describes the isolated macromolecules?

- A. Consist of approximately 90% protein
- B. Highly positively charged molecules that results in considerable electrorepulsion
- C. Contain long branched sugar units
- D. Use linker proteins to form aggregates with hyaluronic acid as central strand
- E. Assembled to proteoglycan aggregates entirely in the endoplasmic reticulum

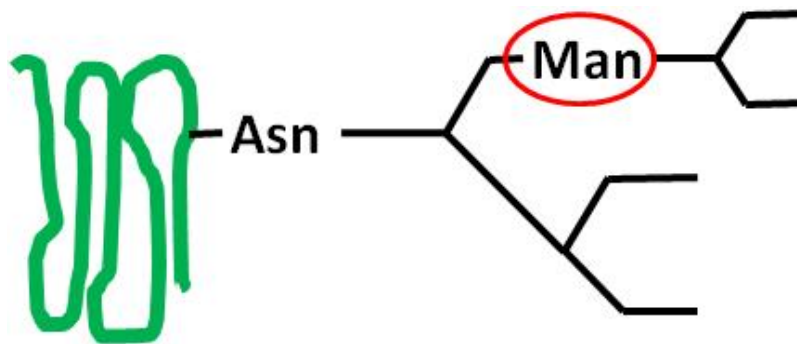
Question #: 3

A 2-month-old girl is brought to the physician by her parents because of a delay in developmental milestones. Laboratory studies show a defect in sorting the enzymes to the lysosomes. Additional studies show a defect in the modification of the lysosomal enzyme as

shown in the figure. Which of the following modifications of the sugar is most likely required for appropriate sorting of this enzyme?

- A. Acetylation
- B. Farnesylation
- C. Oxidation
- D. Phosphorylation
- E. Ubiquitination
- F. Hydroxylation

Attachment:



58628.JPG

Question #: 4

A 20-year-old woman comes to the physician because of a 3-day history of bleeding gums while brushing her teeth. She has been on a special diet for the past three weeks with foods low in carbohydrates and deficient in fruits and vegetables. She does not take any supplements. Which of the following enzymes is most likely lacking its coenzyme and is responsible for her symptoms?

- A. Lysyl oxidase
- B. Prolyl hydroxylase
- C. γ -glutamyl carboxylase
- D. Transketolase
- E. Glucose 6-phosphate dehydrogenase

Question #: 5

A male newborn delivered at full-term is being evaluated for bilateral hip dislocation. Physical examination shows velvety skin and increased joint mobility. Laboratory studies show abnormal

processing of collagen in the extracellular matrix. Which of the following processes is most likely affected in this patient?

- A. Hydroxylation of collagen
- B. Hydrogen bonding in the triple helix
- C. Glycosylation of collagen
- D. Formation of procollagen triple helix
- E. Cleavage of procollagen

Question #: 6

An investigator uses human cell culture to study the role of enhancer proteins in transcriptional regulation. They note that several classes of trans-acting factor proteins coordinate to influence gene expression in this eukaryotic system. Which of the following statements best describes the target of enhancer proteins that function in this type of regulation?

- A. An mRNA sequence
- B. A DNA sequence
- C. RNA polymerase II
- D. The ribosome
- E. The RISC complex
- F. The first stem-loop of an miRNA
- G. A tRNA
- H. An rRNA
- I. DNA gyrase
- J. DNA helicase
- K. DNA ligase

Question #: 7

A researcher studying E. coli lactose metabolism finds that a main regulatory aspect involves a polycistronic mRNA. Gene expression studies show that this operon is induced only under the appropriate environmental and metabolic conditions. Which of the following best explains the proper function of this gene regulatory system?

- A. The inducer of the *lac* operon is cAMP
 - B. CAP/CRP protein only binds to the promoter region in the presence of glucose
 - C. The repressor protein LacI, only binds the operator in the absence of lactose
 - D. A non-functional mutation of the operator will prevent gene expression
 - E. The *lac* operon is only expressed in the presence of both lactose and glucose
-

Question #: 8

A newly married couple comes to the physician for advice because the man has Phenylketonuria (PKU), which has been treated with life-long dietary restriction. The woman is healthy, and their relationship is not consanguineous. In terms of the standard nomenclature for human recessive and dominant disorders, which of the following will be the most likely genotype of their children?

- A. AA
 - B. Aa
 - C. aa
 - D. 50% of their children will be AA and 50% will be Aa
 - E. May not be predicted from the information in the question
 - F. Disorder is multifactorial, so genotype information is irrelevant in this case
-

Question #: 9

A scientist studying an autosomal recessive trait accesses data from publicly deposited genomic repositories to sample DNA sequences from over 250,000 individuals. A specific allele is found to have a frequency of approximately 1 in 100 in the general population. Based on these findings, which of the following is the best estimate of the frequency of homozygotes for this allele in the population?

- A. 1 in 250
- B. 1 in 5
- C. 1 in 50
- D. 1 in 500
- E. 1 in 10,000
- F. 1 in 100
- G. 1 in 250,000

Question #: 10

A researcher studying Marfan syndrome finds that alleles of *FBN1* that are causative of the disorder are found at a frequency of approximately 1 in 10,000. Based on the data, which of the following best describes the incidence of the disorder?

- A. 1 in 2,500
- B. 1 in 5,000
- C. 1 in 10,000
- D. 1 in 20,000
- E. 1 in 25

Question #: 11

An investigator is comparing the incidence of genetic disorders in the human population and selects only those disorders that have an allele frequency of 1 in 1,000. From these specific disorders which of the following would most likely be the rarest type of disorder expected in the population (meaning the least frequently observed)?

- A. A disorder caused by mitochondrial inheritance
- B. X-linked disorders
- C. Autosomal dominant disorders
- D. Autosomal recessive disorders
- E. Multifactorial disorder

Question #: 12

A 20-year-old woman comes to the physician for genetic counseling because she is a known carrier of an allele associated with a rare autosomal recessive form of dystonia. Because it is rare, and challenging to diagnose, the disease incidence is difficult to estimate, however, it is thought to be approximately 1/250,000. What is the estimated risk that this woman will marry a carrier from the general population?

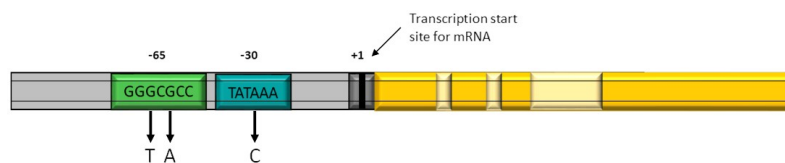
- A. 1/25
- B. 1/50
- C. 1/500
- D. 1/250
- E. 1/5000

Question #: 13

A 3-year-old boy is brought to the physician for a follow-up examination after being diagnosed with beta-thalassemia. Genetic analysis shows that he has three different nucleotide substitution mutations within the promoter of the β -globin gene as shown by the vertical arrows in the figure. In vitro studies show that the mutations lead to reduced expression of the β -globin mRNA. Which of the following cellular events is most likely to be directly disrupted by these mutations?

- A. Coordination of ribosomes binding to mRNA
- B. Stimulation of poly(A) polymerase activity
- C. Binding of basal transcription machinery to promoter
- D. Activation of histone acetylase activity
- E. Appropriate splicing of mRNA by snURPs

Attachment:



170439 ESoft ID.jpg

Question #: 14

A 58-year-old woman with a diagnosis of breast cancer comes to a follow-up appointment with oncologist. The physician asks the patient for consent to perform a biopsy of the tumorous tissue to analyze the gene expression profile in order to guide treatment decisions. The analysis will include the identification of microRNAs (miRNAs) which have a strong role in gene regulation. Which of the following best describes the function of miRNAs in gene expression?

- A. They encode proteins that directly regulate gene expression
- B. They bind to DNA to prevent transcription
- C. They bind to target mRNA to inhibit translation
- D. They are involved in the splicing of pre-mRNA
- E. They act as enzymes to catalyze biochemical reactions

Question #: 15

A 75-year-old man comes to the emergency department because of a 2-week history of lower abdominal pain and severe dysuria. He is diagnosed with a severe urinary tract infection and is started on constant intravenous infusion of an antibiotic at a rate of 10 mg/min. The antibiotic has a half-life of 16 hours, a V_d of 12 L, and a clearance of 0.5 L/min. Which of the following will most likely be the plasma concentration of the drug at steady-state?

- A. 0.1 mg/L
- B. 0.2 mg/L
- C. 20 mg/L
- D. 50 mg/L
- E. 80 mg/L

Question #: 16

A 43-year-old woman is being treated for a urinary tract infection. For the past two days she has been receiving a continuous intravenous infusion of an antibiotic. The antibiotic follows first-order kinetics of elimination. The plasma concentration of the antibiotic has reached steady-state. On the third day of admission, a nurse misreads the physician's instructions and reduces the infusion rate by half. Which of the following will most likely be the consequence of this change in the infusion rate?

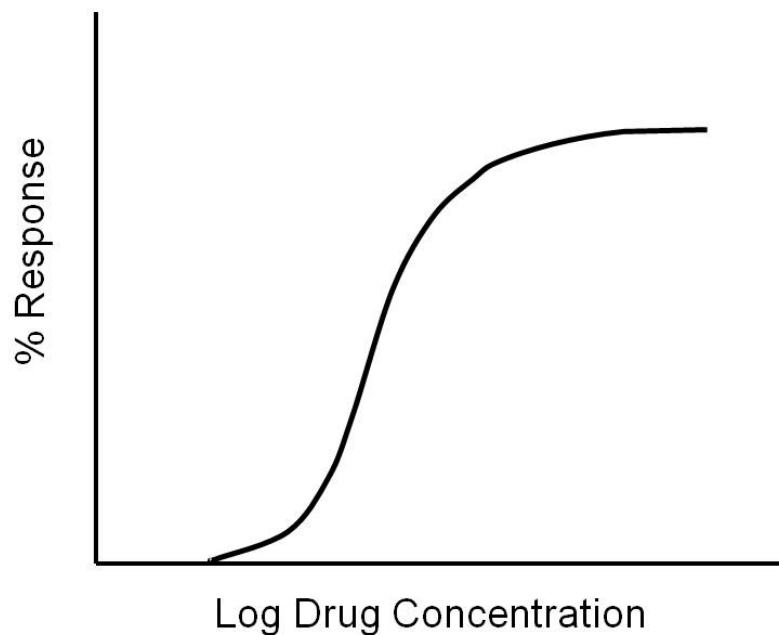
- A. The plasma antibiotic steady state concentration will increase 100%
- B. The plasma antibiotic steady state concentration will decrease 50%
- C. The half-life of the antibiotic will decrease 50%
- D. The half-life of the antibiotic will increase 100%
- E. The clearance of the antibiotic will increase 100%
- F. The clearance of the antibiotic will decrease 50%

Question #: 17

A pharmacologist is studying the properties of a novel beta 2 agonist. He determines the degree of smooth muscle contraction evoked by the drug using an in vitro preparation of porcine airway tissue. The picture shows the dose response curve. Which of the following can most likely be obtained from this experimental curve?

- A. K_d
- B. EC_{50}
- C. ED_{50}
- D. Therapeutic index
- E. Therapeutic window
- F. Bioavailability

Attachment:



BASICS 118.jpg

Question #: 18

A 56-year-old woman is administered a 200-mg dose of an analgesic drug by intravenous bolus injection prior to a surgical procedure. The drug follows first-order kinetics of elimination with a volume of distribution of 120 L and a clearance

of 14 L/h. Which of the following is the approximate percentage of the drug dose to be eliminated in 24 hours?

- A. 25
- B. 50
- C. 75
- D. 88
- E. 94
- F. 97
- G. 99

Question #: 19

The pharmacodynamic properties of a novel antihypertensive drug, Drug X, is being studied in an in vitro preparation of rat renal blood vessels. The studies show that Drug X increases cAMP levels via activation of dopamine D1 receptors. Further studies show that drug X dose-dependently decreases the increase in cAMP levels evoked by dopamine, a full agonist at D1 receptors. Which of the following best describes Drug X?

- A. Physiologic antagonist
- B. Full agonist
- C. Partial agonist
- D. Inverse agonist
- E. Non-competitive antagonist

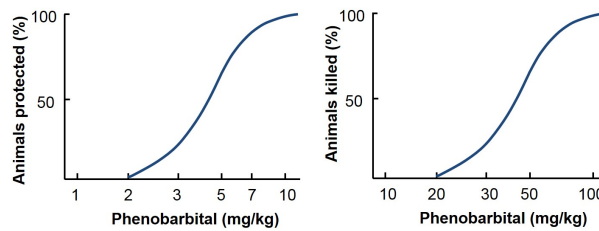
Question #: 20

A team of pharmacologists is determining the quantal dose-response curve for the protective effect of phenobarbital against convulsions in a group of rats. The figure shows the results of the experiment. Which of the following parameters of this drug is most likely obtained from such a dose-response curve?

- A. The therapeutic index
- B. The bioavailability
- C. The EC₅₀

- D. The K_d
- E. The E_{max}

Attachment:



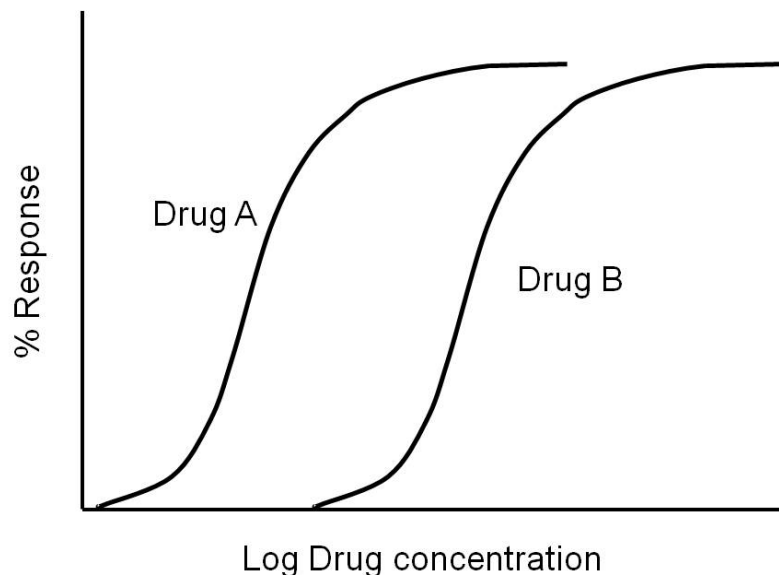
BASICS TI.jpg

Question #: 21

A pharmacologist is studying the properties of two novel drugs, Drug A and Drug B. Both drugs bind selectively and with high affinity to beta-1 adrenergic receptors. The graph shows the effect of drugs A and B on the contractile force of an in vitro preparation of guinea pig cardiac smooth muscle. Which of the following best describes the properties of these drugs?

- A. Drug A has the same potency as Drug B
- B. Drug A is more potent than Drug B
- C. Drug B is more potent than Drug A
- D. Drug A is less efficacious than Drug B
- E. Drug A is more efficacious than Drug B
- F. Drug A is a partial agonist

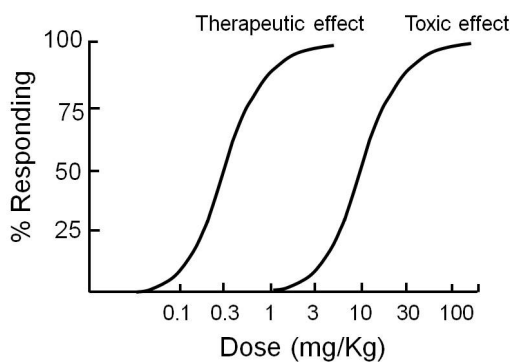
Attachment:



Question #: 22

The effects of a novel drug were tested on a population of mice with experimentally-induced cardiac failure. The therapeutic effect was a 15% decrease in blood volume. The toxic effect was a 30% increase in heart rate. From the results obtained, the ED₅₀ and TD₅₀ were determined. The figure shows the quantal dose-response curves. Which of the following statements best describes the results of the study?

- A. The therapeutic index is 33
- B. The therapeutic index is 10
- C. The ED₅₀ is 1 mg/kg
- D. The TD₅₀ is 30 mg/kg
- E. The therapeutic window is 1 - 30 mg/Kg

Attachment:

BASICS 92.jpg

Question #: 23

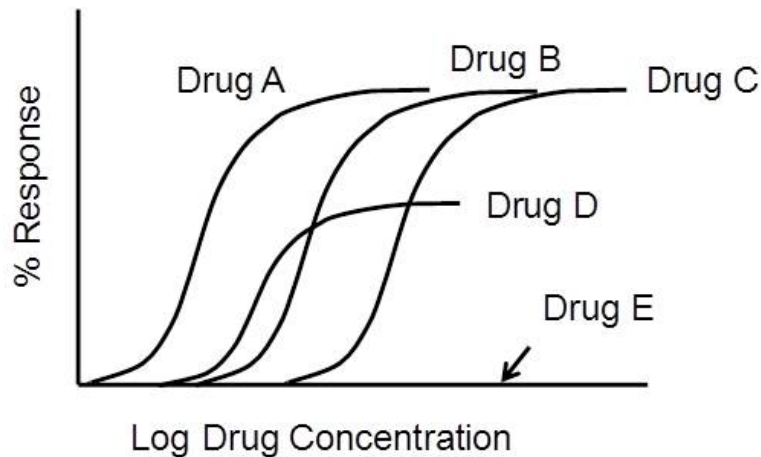
The effects of five new drugs on gastrointestinal smooth muscle contraction are being studied. The five drugs show high affinity for the M3 muscarinic receptor. The figure shows the experimental results. Which of the following drugs is most likely an antagonist?

- A. Drug A
- B. Drug B
- C. Drug C

D. Drug D

E. Drug E

Attachment:



BASICS 84.jpg

Question #: 24

A team of pharmacologists is studying the pharmacokinetic properties of a novel antihypertensive drug. They desire to estimate the bioavailability of the drug after oral administration. Which of the following is most likely needed to determine their desire?

- A. Area under the plasma concentration-time curve (AUC)
- B. Peak plasma concentration
- C. Half-life
- D. Time to peak plasma concentration
- E. Volume of distribution

Question #: 25

A 45-year-old man is brought to the emergency department by ambulance 2 hours after ingesting an overdose of an opioid analgesic. His blood pressure is 102/68 mm Hg, pulse is 56/min, and respirations are 11/min. Physical examination shows a semicomatose man with cool, damp skin and pinpoint pupils. Laboratory studies show that the plasma concentration of the drug is 2 mg/L. The drug follows first-order kinetics of elimination with a half-life of 3 hours. Which of the following is most likely to be the time required for the plasma drug concentration to reach a safe concentration of 0.5 mg/L?

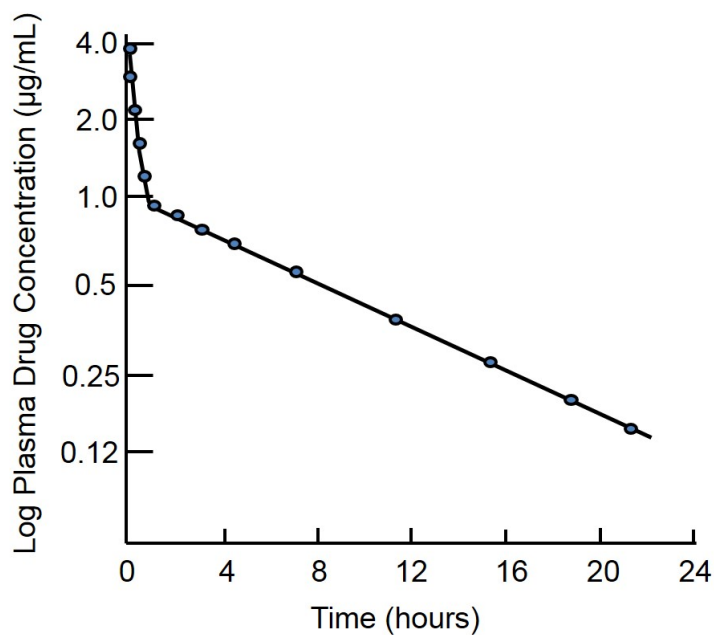
- A. 3 hours
- B. 4 hours
- C. 6 hours
- D. 9 hours
- E. 10 hours

Question #: 26

A 200 mg dose of a new drug was administered as an intravenous bolus to 50 healthy volunteers, and blood samples were analyzed at intervals. The graph shows the average plasma concentration over time. Which of the following values is the best approximation of the calculated volume of distribution of the drug?

- A. 0.5 L
- B. 1 L
- C. 50 L
- D. 100 L
- E. 200 L

Attachment:



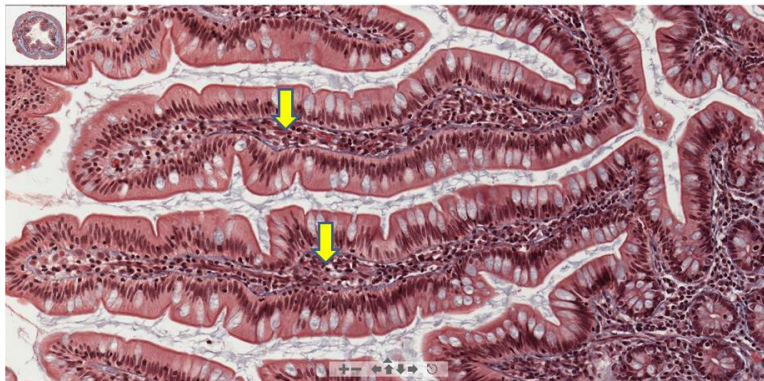
BASICS 82.jpg

Question #: 27

A 28-year-old woman is brought to the physician because of an 8-day history of abdominal pain, and weakness, and a 3-month history of intermittent nausea, vomiting, and weight loss of 6.8 kgs (15 lbs). Physical examination shows an emaciated woman with dried oral mucosa and skin. An endoscopy shows inflammation of the small intestine. A biopsy is shown in the attached image. Which of the following best describes the function of the connective tissue indicated at the tip of the arrows?

- A. Imparts elasticity to the tissue
- B. Withstands unidirectional stress
- C. Serves as energy reserve
- D. Protection from invading organisms
- E. Allows large volume changes in organs

Attachment:



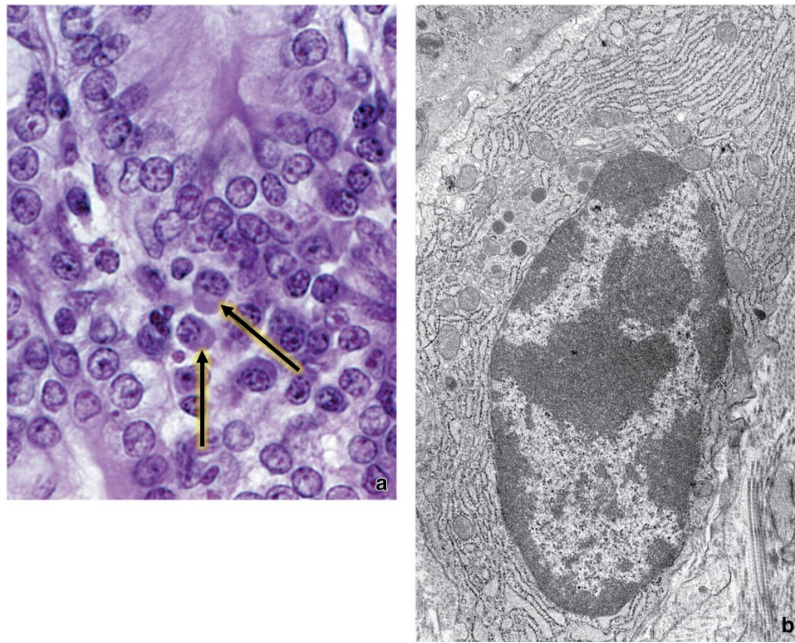
Picture12.JPG

Question #: 28

A 52-year-old woman comes to the physician because of a 3-month history of pain in her lower back and hips, and unintentional weight loss of 6.3 kgs (14 lbs). **Her vital signs are within normal limits.** Physical examination shows tenderness over her hips. Laboratory studies show hypercalcemia (elevated serum calcium levels) and thrombocytopenia (low platelet count). A bone marrow biopsy shows over proliferation of the cell shown in the attached images. Which of the following best characterizes a function of this cell?

- A. Secretion of connective tissue matrix
- B. Production of antibodies
- C. Secretion of elastic fibers
- D. Production of reticular fibers
- E. Synthesis of collagen fibers
- F. Thermogenesis

Attachment:



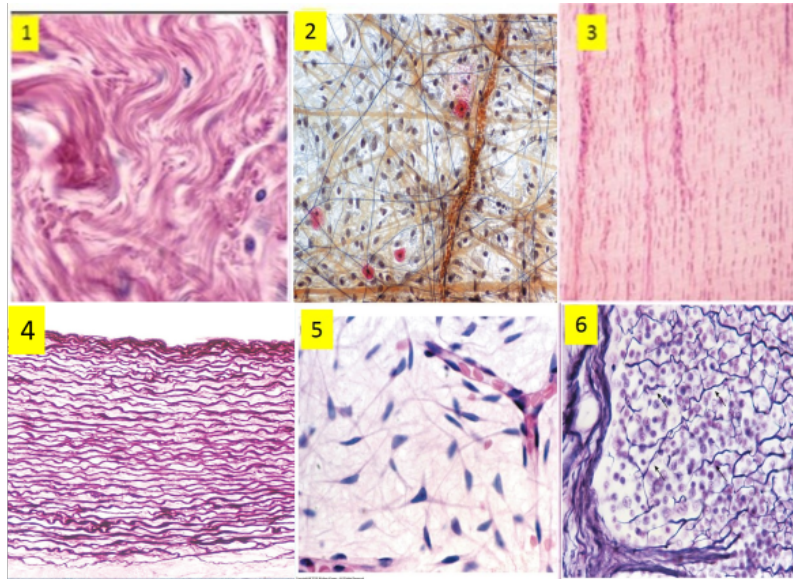
PlasmaCell.jpg

Question #: 29

A 33-year-old woman comes to the emergency department because of a 48-hour history of absent fetal movements. She is at 37-weeks gestation. Physical examination shows absent fetal heart rate. She subsequently delivers a stillbirth fetus. A biopsy of the umbilical cord shows normal blood vessels surrounded by connective tissue. Which of the following numbered images best represents the connective tissue type in the biopsied structure?

- A. 1
- B. 2
- C. 3
- D. 4
- E. 5
- F. 6

Attachment:



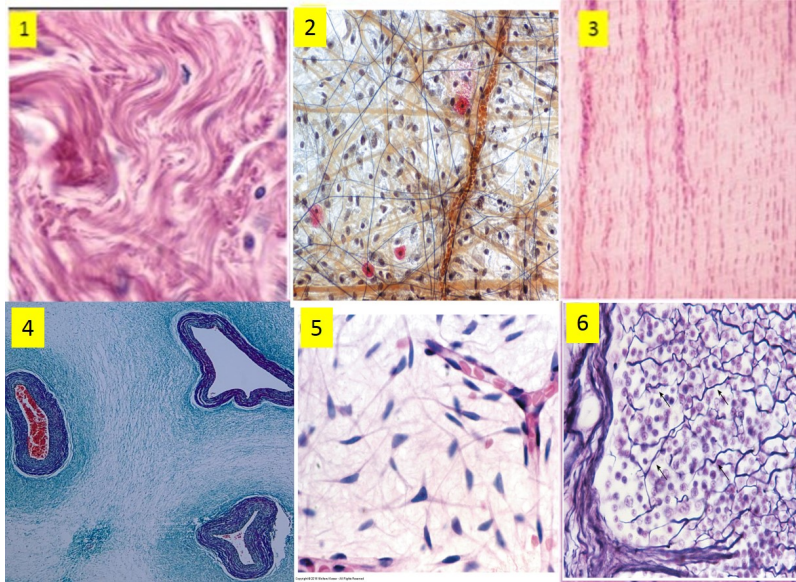
Connective Tissue collage1red1.png

Question #: 30

A 43-year-old man comes to the physician because of a 1-month history of a swelling on his right wrist. Physical examination shows a firm, non-tender mass around the tendon of the flexor carpi ulnaris (forearm muscle). A biopsy shows a mass with infiltration into the dense connective tissue of the skin. Which of the following images best features the histological appearance of the infiltrated tissue?

- A. 1
- B. 2
- C. 3
- D. 4
- E. 5
- F. 6

Attachment:



connective tissue collage .jpg